

A2 FILE CMD

User guide · 0.7.6

6502 and 65C02 · BOOT / EXTRA / XL

```

/A2XL65C02
Name#      Type  Aux  Size
-----
<UP>
A2FILE/    <DIR>      4
DEMO/      <DIR>      2
IMGHGR/    <DIR>      1
A2FILE.SYS SYS $0000    6043
BASIC.SYS  SYS $0000   10240
PRODOS     SYS $0000   17128

/A2XL65C02/DEMO
Name#      Type  Aux  Size
-----
<UP>
DHGR.RAW   BIN $0000   16384
DHGR.RLE   BIN $0000   12030
DOS33.DSK  BIN $0000  143360
HELLO      BAS $0801     98
HGR.RAW    BIN $0000    8192
HGR.RLE    BIN $0000    978
LETTER     AWP $0000   1336
README     TXT $0000   1315
SAMPLE     TXT $0000    712
SAMPLE.BNY BIN $0000   1152
SAMPLE.SHK BIN $0000    985
TINY.2MG   BIN $0000   32832
TINY.PO    BIN $0000   32768
WELCOME.MB BIN $0000     56

A2 FILE CMD 0.7.6      64383 of 65535 blocks free
Parent directory

TABPanel RET Open SP Tag C Copy V Move R Ren D Del K Mkdir ! More ? Help

```

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English guide · Generated 10 September 2026

Version 0.7.6 — A two-panel ProDOS file manager for an Apple II with 128 KB and 80-column support. Free software by Arnaud Verhille, under GPL v3.

Start here

Choose **6502** for an original, unenhanced IIe. Choose **65C02** for an enhanced IIe or IIc; this edition also supports an optional AppleMouse II. The IIgs has not been tested. The launcher checks the CPU and memory before starting and identifies the edition on its title screen.

Choose your disks

Edition	What to use
BOOT + EXTRA	Two 140 KB floppies for the same CPU and release. Boot BOOT; EXTRA supplies the additional tools and BASIC.SYSTEM.
XL	One bootable 32 MB .2mg with all 44 overlays, BASIC.SYSTEM and demonstration files. No EXTRA disk is needed.

The download names include the CPU, role and version:

CPU	Image names
6502	A2FILECMD-6502-B00T-0.7.6.dsk, A2FILECMD-6502-EXTRA-0.7.6.dsk, A2FILECMD-6502-XL-0.7.6.2mg
65C02	A2FILECMD-65C02-B00T-0.7.6.dsk, A2FILECMD-65C02-EXTRA-0.7.6.dsk, A2FILECMD-65C02-XL-0.7.6.2mg

BOOT and EXTRA are supplied as .dsk in DOS sector order; XL uses .2mg. Use the downloaded files directly: changing an extension does not convert an image. Each floppy image is 143,360 bytes. Check downloads against SHA256SUMS-0.7.6.txt; if all release files are together, run:

```
sha256sum -c SHA256SUMS-0.7.6.txt
```

Boot the image, or launch A2FILE.SYSTEM from a ProDOS selector. To install elsewhere, keep A2FILE.SYSTEM beside its complete A2FILE/ directory. Do not mix A2FILE.CODE and native plugins from different builds.

Read the panels

The highlighted panel is active; **TAB** switches sides. Each panel shows names, file types, auxiliary types and sizes. The selected row is highlighted, a star marks a tagged file, and **L** marks a locked file. The header's star identifies the sort order. Below the panels are free space, selection details, messages and the key bar. **?** opens the help screen.

The program remembers panel directories, sorting and the active side in A2FILE/A2FILE.CFG when you quit. On the first XL start, the right panel opens DEM0/; try its text, pictures, music and sample archives.

The companion floppy and disk swaps

EXTRA contains 25 tools absent from BOOT, plus the common menu and BASIC.SYSTEM. Use the **same CPU and version** on both disks.

With two Disk II drives, keep BOOT in **slot 6, drive 1** and put EXTRA in **slot 6, drive 2**. The **!** menu lists tools from both disks.

With one drive, choose the tool normally. If a disk is missing, the prompt names the required volume, slot, drive and file. Press **1** or **2** to choose the drive, insert the named disk, then press **Return**. If the input file was on the removed disk, a second prompt asks for that disk. **Escape** cancels and returns to the panels. The chosen drive is remembered for the session; these plugin-loading prompts use slot 6.

CPU	Volume names shown in swap prompts
6502	BOOT /A2FC6502; EXTRA /A2EXTRA6502; XL /A2XL6502
65C02	BOOT /A2FC65C02; EXTRA /A2EXTRA65C02; XL /A2XL65C02

The menu retains all 43 commands when EXTRA is absent. Tools that need both source and destination online still require another drive or volume. **DISKCMP S** and **W** → **Copy** have their own single-drive exchange modes.

Protect files in /RAM

Copy anything important out of /RAM before displaying DHGR, loading music, extracting ShrinkIt, using disk-image operations or physically formatting a Disk II floppy. These operations use auxiliary memory and rebuild /RAM empty; the message line reports it. Plain HGR pictures do not clear /RAM. Avoid writing to /RAM while music is playing.

Keys

Key	Action
Up / Down	Previous / next entry.
Left / Right, < / >, - / +	Previous / next page; [/] jumps to first / last entry.
TAB / =	Switch panel / show the same directory in the other panel.
RETURN / ESC / /	Open / go up / list volumes. Return asks before running a program.
', then letter or digit	Jump to the next name with that initial.
SPACE / *	Toggle the selection's tag / invert all tags.
Ctrl-T / Ctrl-N / Ctrl-R	Tag all / untag all / reread both panels.
S / M	Change sorting / tag files absent or different in size/date in the other panel.
C / V	Copy / move tagged entries, otherwise the selection, to the other panel. Includes directories. Move deletes originals after copying.
D	Delete tagged entries or the selection, including directory contents, after confirmation.
R / K	Rename / create a directory. Names: letter first, then letters, digits or periods; maximum 15 characters.
A / L	Change hexadecimal type/auxtype / lock or unlock. Locked files refuse deletion and renaming.
T / H / I	Read text / hexadecimal / picture. T also lists BAS and AWP files.
E	Edit text; on a directory or . . ., create a text file.
P / X	Pause/resume music / run a program after confirmation, replacing A2FC.
W / F / !	Disk-image operations / format / plugin menu.
? / 1 ... 0 / Q	Help / key-bar buttons / quit to ProDOS after confirmation.

Copying preserves name, type and auxiliary type. For an existing target, choose **O** overwrite, **S** skip, **A** overwrite all or **N** overwrite none. Existing destination directories are filled in. Copying into the same or a nested source directory is refused.

Long operations display progress. **ESC** interrupts; completed work remains, an incomplete ordinary file copy is removed, and the final message reports what was done. Read the result before removing a disk.

The mouse

On 65C02, an AppleMouse II supplements the keyboard. Click a row to select it; click the selected row again to open it. Click a path to go up, a column header to change sorting, or a key-bar button to invoke it. Viewers, the editor and prompts use the keyboard.

More tools in the ! menu

Select the item first, press **!**, then choose a tool. **Up/Down** moves a line, **Left/Right** changes page, and a letter jumps to a matching initial. The following tools supplement the main keys and readers.

On BOOT and XL

Tool	Operation
COMPARE	Compare the selected file byte by byte with the same name in the other panel.
TXTCONV	C = CR, L = LF, D = CRLF, H = clear high bit, S = set it, T = expand tabs, A = transliterate UTF-8 accents. Write in place or to the other panel.
DATE	S sets date/time from DDMMYYYYHHMM (1940–2039); impossible dates are rejected. F stamps modification dates on tagged files or the selection. Creation dates stay unchanged; a hardware clock may replace the entered time.
VERIFY	Read tagged files (skip directories), the selection, or every block of a volume. Report processed files and errors; ESC cancels. No writes.
TAGPAT	Name patterns: = any string, ? one character. Add comma-separated filters: T04 TXT, >2000 or <2000 bytes, D modified today. T tags, U untags, X replaces tags.
VOLINFO	Audit allocation and fragmentation. M = bitmap (. free, # used), F = selected file blocks, E = export to the other panel. N/P pages; ESC returns. No repairs.
VOLNAME	Rename a ProDOS volume and update the affected panel/program paths.
WIPE	F zeroes free blocks after confirmation. W zeroes the whole volume after ERASE; the running program's volume is refused.

On EXTRA and XL

Tool	Operation
SEARCH	Find text in the active directory and tag matching files, ignoring case.
FIXTYPES	Set type/auxtype from suffixes on tagged files or the selection; optionally remove suffixes. Image suffixes and .SYSTEM stay.
GOTO	Nine favourite directories: A adds, D then a digit removes, 1–9 jumps. Saved in A2FILE/GOTO.CFG.

Tool	Operation
FIND	Search the volume by name pattern; start with " to search contents, ignoring case. TAB sets type (T, two hex digits) and modification dates (D, inclusive YYYYMMDD, 1940–2039); A clears filters. Undated files are excluded by date filters. N shows the next 20 results; Return jumps there. V on a text result shows occurrence offsets (hex) and excerpts; N/Space continues, ESC returns.
BLKVIEW	Read device or image blocks: H hex/ASCII, D directory, I index, N/P block, Space page, G four-digit hex block, F find four bytes (8 hex digits), A find next, X extract blocks, ESC back. Source stays unchanged.
DISASM	Read BIN/SYS as assembly: N/Space next, P previous (last 64 pages), C 6502/65C02, G six-digit hex file offset, L four-digit load address, R start, E export, ESC back. BIN uses its auxtype; SYS starts at \$2000.
CRC	Calculate CRC-32 for the selection or tagged files. Results appear in pages of 20; a key continues, ESC at a page boundary stops the batch.
IDENT	Identify a file by content; text statistics cover its first 512 bytes.
MDVIEW	Read Markdown or long text with wrapping, headings, lists and code; page forward/back.
RENAME	Batch prefix, suffix, extension replacement/removal or numbering. For example E then BAK sets .BAK. Conflicts are skipped.
IMGCONV	Convert PO/HDV, DSK/DO and 2MG into the other panel, preserving disk blocks.
BOOTBLK	Copy ProDOS boot blocks from the boot volume to another volume after confirmation.
UNDELETE	Browse deleted ProDOS entries. N skips; R recovers a validated candidate to another online volume. Existing names are refused.
DISKCMP	V compares online ProDOS volumes; I compares images; S compares two Disk II disks on one drive. Reports differing blocks and the first mismatch.
MKIMAGE	Create an empty ProDOS PO or 2MG: 140 KB, 800 KB, 2/4/8 MB or 32,767 blocks. New images are data volumes, without a boot program.
RESCUE	F recovers a file; V recovers a ProDOS volume. Uses up to 30 attempts per block, zero-fills unreadable chunks and writes a LOG. Destination must be another online volume.
SYNC	Recursively copy missing or newer files to the other panel after confirming direction. Destination-only files remain; copies are read back before replacement.
TREE	Show file sizes and cumulative directory totals. Space advances a page; ESC exits.

Recovery and comparison limits

UNDELETE never changes the source directory, indexes or bitmap. It checks retained pointers, block counts, free blocks and index halves swapped by ProDOS DESTROY. An interrupted DESTROY can leave a deleted entry with damaged allocation information: the deleted marker alone is insufficient. Reused, inconsistent or ambiguous candidates are refused. Standard seedling, sapling and tree files, including sparse files, are supported; deleted directories and resource forks are not. Inspect recovered data before relying on it.

RESCUE writes BASE.REC for a file. Whole-disk recovery writes raw ProDOS parts BASE.P01, BASE.P02, etc., up to 16,000 blocks each. Concatenate them in numeric order for a PO image; a single part can simply receive .P0. BASE.LOG records zero-filled chunks and completion or interruption. Partial output stays after cancellation or a write failure.

SYNC skips equal-date or newer destination files. A source with an unknown date does not replace an existing file. It uses A2FC.SYNC temporarily and A2FC.BAK for rollback, preserving pre-existing files with those names. If replacement fails, the original is restored where possible; keep any remaining A2FC.BAK. Overlapping directory trees are refused. SYNC and TREE support paths shorter than 64 bytes and up to 16 directory levels; read errors and unsupported resource forks produce an error/incomplete result.

DISKCMP S buffers two blocks per exchange, so a full comparison needs many swaps. Each prompt names the expected disk and slot/drive; **1/2** changes the drive, Return retries and Escape cancels. A read error or cancellation never produces an “identical” verdict. Image comparison supports PO/HDV, DSK/DO and ProDOS-order 2MG.

VOLINFO supports ProDOS files, both forks and directories up to 16 levels. Large volumes take longer; incomplete counts are unconfirmed. File lists show data, index, master and extended blocks, omitting sparse holes. Exports include the selected file and describe the scan before report creation; existing names are refused. Only complete exports end with **END REPORT**; partial files remain. **MKIMAGE** refuses existing names and removes cancelled/incomplete new images; 32,767 blocks is the maximum that fits in a single ProDOS image file.

DISASM shows file offsets, 16-bit CPU addresses, bytes and instructions. **C** changes decoding at the current offset and resets page history; **65C02** includes Rockwell/WDC extensions. Unknown/truncated instructions appear as **.BYTE**. **E** exports from the current offset to EOF as a new **TXT** file in the other panel, using the displayed CPU and load address. Existing names are refused. **ESC** cancels; errors or cancellation keep partial output. Exporting preserves the current page.

BLKVIEW F searches forward from the current block, including matches across block boundaries; **A** continues without wrapping. **X** extracts from the current block: enter a four-digit hex count (maximum **7FFF**) and a new filename in the other panel. Device sources require another destination volume. Errors retain partial output.

Disk writes and copies automatically read back every written block. A readback error stops at the first failing block. On one drive, each prompt identifies **SOURCE** or **TARGET** copy, the source volume and the slot/drive.

Reading files and pictures

Text, hexadecimal and AppleWorks

Use **T** for text, **H** for hexadecimal. **Space**, **Return** or **Down** advances a page; **B** or **Up** goes back; **ESC** returns to the panels. The text reader clips lines beyond 80 columns and remembers up to 96 page starts. Use **MDVIEW** for wrapped text. The hex reader shows addresses, bytes and their text representation.

T on a **BAS** file lists Applesoft line numbers and keywords. **T** or **Return** on an **AWP** document reads AppleWorks word-processor text; formatting commands are omitted and tabs expanded. AppleWorks databases and spreadsheets are not supported.

Pictures

Return recognizes pictures by content; **I** explicitly tries the picture reader. Supported formats are raw **HGR** (8,192 or 8,184 bytes), raw **DHGR** (16,384 bytes, auxiliary plane first), and **HGRR/DHRR** version 1 **RLE** files. **Left/Right** shows the previous/next picture; any other key returns. Remember that **DHGR** clears **/RAM**.

The Mockingboard music

Return on an **MB1** .MB stream starts playback (maximum 2,304 bytes). **P** pauses/resumes; another music file replaces it. Playback ends at the stream's end, on quit or when another program runs. Disk II reads may pause it briefly. Loading music clears **/RAM**; an absent Mockingboard is reported.

The text editor

E edits a file; on a directory or **. . .**, it creates one. The editor holds up to 8 KB, uses **CR** line endings and strips the high bit on loading. Long lines do not wrap. A star on the status bar means unsaved changes.

Arrows move; **Delete** / **Ctrl-D** erase left / right. **Ctrl-A** / **Ctrl-E** go to line start / end; **Ctrl-P** / **Ctrl-N** changes page; **Ctrl-T** / **Ctrl-B** goes to text start / end. **Return** splits a line; **Tab** inserts four spaces.

ESC opens the menu: **S** save, **X** save and exit, **Q** quit without saving (confirm if changed), **ESC** continue editing.

Saving preserves the file's ProDOS type and auxiliary type.

Disk images and formatting

Disk images

W opens three operations: write the selected image to a formatted target disk, read a disk into a new image in the active directory, or copy one disk to another. Supported containers are PO, DSK/DO and 2MG.

For a one-drive copy, choose the same drive for source and destination, then follow the disk prompts. Writing requires **ERASE** in capitals and refuses the running program's disk. Check the named target before confirming. Escape cancels before writing. Disk-image operations can clear /RAM.

A disk image as a folder

Return on PO, DSK/DO or 2MG opens a supported ProDOS or DOS 3.3 image read-only. Navigate with Return and Escape; Escape at its root leaves the image. **C** extracts tagged files, or the selection, to a real ProDOS directory in the other panel, preserving type and auxiliary type.

ProDOS image extraction supports files up to 128 KB (seedling/sapling). Enter subdirectories and extract their files individually; recursive extraction and writing into an image are not supported.

A real DOS 3.3 disk appears in / as DOS 3.3, with its slot and drive. Return opens its catalog; C extracts files to ProDOS. Applesoft, Integer and binary DOS headers are removed during extraction. DOS 3.3 access is read-only.

Formatting a disk

Press **F** or choose **FORMAT** in **!**. Select the drive, enter a volume name, then type **ERASE** and press Return. The list shows slot, drive, volume and capacity. The program's own volume is marked **IN USE** and cannot be formatted. Escape or a different confirmation word cancels.

Disk II formatting clears /RAM; move its files elsewhere first. It is refused if A2FC itself runs from /RAM. Formatting another block device does not require that extra RAM reset. Music stops when FORMAT opens. The result reports success or an error; Escape returns directly to the panels.

Archives and programs

Unpacking archives

Set the destination in the other panel, select the archive, then use **!**:

Tool	Supported archives
UNSHRINK	ShrinkIt .SHK: stored data, LZW/1 and LZW/2. Files retain ProDOS type and auxiliary type; names are adapted to ProDOS. Disk-image members become PO files. Resource forks and comments are skipped.
BINARY2	Binary II .BNY/.BQY: extracts members with their names and attributes. Directory entries are skipped. Compressed members may need a second extraction with UNSHRINK.

ShrinkIt extraction clears /RAM and refuses it as a destination. Keep archives and recovered files on another volume.

Running an Applesoft program

Return or **X** runs BAS, SYS or BIN after confirmation. The launched program replaces A2FC; it does not automatically return. A BIN loads at its auxiliary address, which must be between \$0800 and \$BAFF.

Applesoft requires BASIC.SYSTEM, supplied on EXTRA and XL. A2FC searches the program's volume, its own volume and the companion. With one floppy drive, keep the BAS program on another online volume so it remains readable after BASIC.SYSTEM loads.

From the Applesoft] prompt, reinsert BOOT if needed and enter:

```
-/A2FC6502/A2FILE.SYSTEM
```

Use /A2FC65C02/ for the enhanced BOOT, /A2XL6502/ or /A2XL65C02/ for XL, or the actual installation path. -A2FILE.SYSTEM also works when the current prefix is its directory. BAS paths over 46 characters use a fallback launch method; use a shorter path if launch fails.

VDrive: two volumes over the serial line

A detected Super Serial Card or //c serial port can expose two remote ProDOS volumes at **115,200 bps**. Slot 2 is tried first. The status line identifies the serial card and assigned drive slots; the remote volumes then work like local ones for browsing and copying.

Use a compatible host such as ADTPro's virtual-drive server, `veserver.py` or `sur1-server`. The host supplies date/time during reads. A missing or disconnected server produces an I/O error. The driver is removed when A2FC quits or launches another program. VDrive is tested in POM2; operation on a real Super Serial Card or //c remains unverified.

Limits and troubleshooting

Symptom or limit	What to do
CPU or memory refusal at startup	Use the matching CPU build, with 128 KB and 80-column support.
Missing or stale plugin	Insert matching BOOT/EXTRA from the same release. Keep A2FILE.CODE and its native plugins together.
Disk changed but old contents remain	Press Ctrl-R to reread both panels.
Run failed: file not found	Check the selected program, its path and BASIC.SYSTEM for BAS files.
Image cannot be opened	Check its actual format; renaming PO to DSK does not convert it. Some file/image features are unsupported.
Directory exceeds 139 entries	A2FC uses windows in disk order, without sorting. Cross a window edge to continue.
Very deep directory copy refused	Simplify the tree or copy smaller subdirectories; recursive operations have bounded working space.
Incomplete scan or recovery	Read the reported limits or errors; inspect recovered data and keep originals/backups.

For building and plugin development, see the project README, `sdk/README.md` and `bench/README.md`. make disk builds both CPU families; make test runs host checks. Planned features and remaining limits are in `TOD0.md`.